

Test Case 1b: Joukowski Airfoil

- **Validation of steady CFD analysis, required**

- <https://github.com/Drag-Prediction-Workshop/DPW8-Scatter/blob/main/TestCase1b/Joukowski.pdf>

- **Settings**

- Steady CFD RANS **French Vanilla SA-[neg] (All terms!)**
 - Adiabatic Wall (not isothermal)
 - Characteristic Farfield (**100 chords away**)
 - Use periodic boundary conditions for sidewall boundary conditions
 - Converge residuals to machine precision ($\sim 1e-10$)

- **Grids**

- Committee-supplied grid family

- **Conditions**

Mach	Re _c	T _{static}	α	γ	Pr	Pr _t	Farfield $\chi = \tilde{\nu}/\nu$
0.15	6×10^6	520.0 R	0.0°	1.4	0.72	0.9	3

- **Sutherland's Law**

$$\mu(T) = \mu_0 \left(\frac{T}{T_0} \right)^{3/2} \left(\frac{T_0 + S}{T + S} \right) \quad \begin{matrix} \mu_0 = 1.716 \times 10^{-5} \frac{\text{kg}}{\text{m s}} \\ T_0 = 491.6^\circ \text{R} \\ S = 198.6^\circ \text{R} \end{matrix} \quad \frac{\mu(T)}{\mu_{ref}} = \left(\frac{T}{T_{ref}} \right)^{3/2} \left(\frac{1 + S/T_{fef}}{T/T_{fef} + S/T_{fef}} \right)$$